Structure-Pain Discordance and 48-Month Knee Pain Prediction Error in the Osteoarthritis Initiative

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Abstract

Background: There exists inconsistency between the radiographic structure severity and the patient self-report pain in the context of knee osteoarthritis. Whether this disagreement independently predicts future unpredictability in pain reports was not established by any directly matching study identified in the literature reviewed. **Objective:** Herein, we aim to develop and internally validate a 48-month Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain prediction model, assess the predefined discordance effect size, and explore the chance for uncertainty-based selective deferral. **Methods:** Our secondary data analysis used data from the multisite Osteoarthritis Initiative (OAI). Of the initial 8,920 eligible knees of 4,490 subjects, 8,232 knees of 4,146 subjects possessed 48-month follow-up WOMAC pain scores. The expected baseline pain score was determined based on the baseline Kellgren-Lawrence (KL) grade within training folds, and discordance was measured as the absolute difference between the observed and expected pain scores. An elastic net model was trained using participant-grouped nested five-fold cross-validation. **Results:** Held-out mean absolute error (MAE) equaled 1.787 WOMAC points (95% confidence interval [CI], 1.741 to 1.835). In our frozen unadjusted primary analysis, an increase of one WOMAC unit in the absolute discordance value was related to an additional 0.301 WOMAC unit absolute error (95% CI, 0.275 to 0.326; $p = 8.18\times10^{-118}$). Yet, there existed considerable correlation between the absolute discordance and baseline WOMAC pain scores ($r = 0.751$). Post-freeze sensitivity analyses controlling for baseline pain revealed that the discordance association did not persist (absolute discordance coefficient = -0.011 WOMAC units; 95% CI, -0.043 to 0.022; $p = 0.52$). Estimated uncertainty positively correlated with absolute error ($\rho = 0.573$; $p < 0.001$). By deferring the 823 most uncertain knees (10.0%), we achieved a reduction in mean absolute error in the retained cases from 1.787 to 1.582, while rates of subgroup abstention varied from 2.9% to 39.5%. **Conclusions:** Discordance showed strong tracking of prediction errors in our frozen unadjusted primary analysis, yet this association did not persist when we controlled for baseline symptoms, suggesting the baseline symptomatology better explains the discordance error correlation than does discordance itself. Discordance was indeed a predictive feature of our uncertainty-based abstention strategy, which therefore cannot be considered independent verification of discordance effects. The observational nature of our study limits inference of causality. **Registration:** The study was not registered in a public registry; an internal dated preregistration lock was created before the primary test.

Keywords: knee osteoarthritis; WOMAC pain; structure-pain discordance; machine learning; uncertainty; selective prediction; abstention

Introduction

Osteoarthritis involves multiple tissues within joints. Cartilage, subchondral bone, synovium, ligaments, tendons, and, in particular regarding the knee joint, the meniscus all may play roles. From a clinical perspective, OA presents with pain, stiffness, reduced range of motion, and disability. However, the burden experienced by affected patients is quite heterogeneous. Known risk factors for OA include

older age, obesity, and joint injuries/surgeries (National Institute of Arthritis and Musculoskeletal and Skin Diseases).

Plain radiography is commonly performed to diagnose structural changes of knee OA. Severity of such alterations is assessed using the Kellgren-Lawrence grade, which ranges from 0, indicating absence of radiographic OA, to 4, reflecting more structurally advanced disease. The disconnect between structure severity and pain is well established; there are patients suffering substantial pain in presence of minor radiographic abnormalities as well as others exhibiting marked radiographic changes with relatively little pain. Previous investigations have found such associations real but nontrivial (Finan et al. 363; Bedson and Croft; Neogi et al.). There exist underlying processes contributing to OA-related pain that cannot be fully understood through simple structural imaging alone, processes such as inflammation, trabecular/meniscal alterations, past trauma, nervous system sensitization, psychosocial mechanisms, individual differences, and so forth (Finan et al. 363; Bedson and Croft; Neogi et al.). On the contrary side, highly advanced structural disease does not equal to symptomatic severity. Thus, there exist instances whereby pain cannot easily be predicted using structure.

The manifestation of OA-related symptoms also displays demographic heterogeneity. Within the context of the Osteoarthritis Initiative (OAI), Black patients have been found to report significantly higher WOMAC pain levels compared to Whites in longitudinal follow-up visits (Vina et al. 471). Women, compared to men, appear to have greater pain and disability in OA at equivalent radiographic severity (Segal et al. 1045). Thus, evaluating prediction-system performance across different demographics is necessary when analyzing uncertainties.

Knee pain prediction models exist in the literature. Previous OAI and Foundation for the National Institutes of Health OA studies have utilized statistical models, neural networks, automated machine learning, and other forms of predictive modeling to forecast future knee pain (Halilaj et al. 1643; Guan et al. 363; Chen and Or; Li et al.; Tong et al. 1045; Alkhatatbeh et al.). Knee OA studies incorporating OA pain prediction have recently implemented uncertainty estimation, risk-coverage assessment, and selective forecasting (Gao et al. 1; Yu et al.). Discordance between pain and structure was investigated previously in multi-modal phenotyping and decision-making investigations into OA (Ahadian et al.). Selective forecasting is an extension of prediction that enables models to make predictions only if confidence levels meet a minimum threshold (Kompa et al.; Swaminathan et al. 188). Within the literature reviewed for this study, no directly matching publication was identified that analyzed whether discordant knees could have greater errors during longitudinally-held-out testing, such that a nested uncertainty framework could be utilized to select away from such difficult longitudinal predictions and audit subgroup performance instead.

For our current investigation, the target population was adults with or at high risk of knee osteoarthritis. The main methodologic objective of the current study is to aid future researchers and clinicians building predictive models with understanding of when future pain prediction would likely be less certain. Our predictive model was never intended to be released as a tool for patients or clinical use for diagnostic, therapeutic intervention selection, or automated clinical triage.

Our first hypothesis states that greater absolute discrepancy between KL grade and baseline WOMAC pain would be associated with greater absolute error in prediction of 48-month WOMAC pain. Secondary objectives include: (1) determining if this hypothesis holds for 24- and 36-month WOMAC pain; (2) evaluating whether an uncertainty approach can accurately discriminate future pain predictions resulting in high error prior to seeing results; (3) whether abstaining from high-error

forecasts reduces error within retained forecasts; and (4) comparing subgroup performance of predictions. Adjustment for initial pain was included post hoc in an internal-review sensitivity analysis.

Methods

Study design and data source

In this secondary longitudinal analysis, de-identified data were obtained from the Osteoarthritis Initiative (OAI) through the National Institute of Mental Health Data Archive (NDA) OAI data resource according to relevant repository access and data usage conditions. The OAI is an observational multicenter cohort study in the USA designed to explore the development and progression of knee osteoarthritis (National Institute on Aging). Between 2004 and 2006, a cohort of 4,796 adults between ages 45 to 79 years old were recruited, either presenting symptomatic knee osteoarthritis or at increased risk of developing it (Nevitt et al.). This study made use of data collected at baseline with follow-up visits until 48 months (2008-2010). As OAI represents a research cohort and not a population probability sample, one must not assume representativeness for any other patients or health systems or geographies. The unit of analysis was the knee. Individual participants could provide data on both knees, but appropriate sampling methodology would ensure that no individual had knees in separate partitions of data for modeling, resampling, or validation purposes. The OAI itself is not interventional, so regular care outside of the cohort does not affect inclusion in analyses nor modification of predictive models.

This analysis was internally preregistered on 21 July 2026 before conducting any tests examining the relationship between discordance and prediction error moving forward. The final analytical freeze was declared after completion of all prespecified and secondary analyses. Preparation of manuscripts was not allowed to change the cohort characteristics, model formulations, validation strategy, decision criteria, hypotheses, or numerical outputs generated in analyses.

Cohort construction

Eligibility and exclusion criteria for the parent cohort within the Osteoarthritis Initiative (OAI) have been documented extensively previously (Nevitt et al.). In order to develop prediction models for our current investigation, several key inclusion criteria had to be fulfilled, including having the frozen knee-level analytic cohort; having a central knee radiograph from OAI X-Ray Reading Project 15; a baseline Kellgren-Lawrence (KL) grade; baseline knee-specific WOMAC pain score; and no knee replacements according to the respective OAI replacement status field. We selected Project 15 based on OAI documentation as the appropriate longitudinal reading project pertaining to semi-quantitative radiographic scoring up to 48 months. Radiographs scored by the alternative projects were not merged with Project 15 readings because the same radiograph could potentially have been scored by all projects.

After applying these criteria, the baseline population consisted of 8,920 knees from 4,490 participants. For the main analysis of predictions, 48-month WOMAC pain data was also needed, leaving 8,232 knees from 4,146 participants for analyses. An additional longitudinal secondary cohort needed WOMAC pain at baseline, 24, 36, and 48 months, totaling 7,807 knees from 3,937 participants.

No formal pre-study sample size calculation was performed since the present work uses all eligible knees from the OAI database instead of sampling a fixed population size. Our 8,232 knees main cohort permitted participant-grouped five-fold development and internal evaluation, with each outer fold consisting of 1,643-1,649 knees from 829-830 participants each. No instances were omitted to balance fold sizes.

Outcome, predictors, and structure-pain discordance

The principal endpoint was knee-specific Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) pain score at 48 months. In the WOMAC employed here, pain was assessed separately per knee in five WOMAC pain items, summed into a 0-20 scale subscale in which larger numbers represent more pain (Nevitt et al.; McConnell et al.). The endpoint of 48 months was chosen as a pre-specified period to permit prognostication of symptoms over the first four years, covered by radiographic follow-up in OAI Project 15. Measurement of outcomes occurred as part of the OAI before construction of the current models, so neither participants nor OAI personnel were aware of any predictions made. Pre-specified baseline covariates were baseline WOMAC pain, categorical Kellgren-Lawrence (KL) grade, medial and lateral joint-space-narrowing grade, age, body mass index (BMI), sex, race, Hispanic ethnicity, prior injury of the respective knee, and prior surgical procedure on the respective knee. The choice of covariates was guided by clinical significance, baseline availability, and applicability to answering the structure-pain question; no screening of outcome measures univariate associations occurred. While race was stored as a pre-specified repository covariate, its interpretation was exclusively demographic, and only audit of performance and coverage was permitted. WOMAC pain at 24 and 36 months was reserved for secondary analysis of longitudinal trends and was not used as a baseline covariate.

The KL and joint-space-narrowing grades came from centralized reading of radiographs in OAI Project 15. Documentation by the OAI states that paired longitudinal radiographs were blindly assessed, resolving disputes between readers via an arbitration process (Osteoarthritis Initiative 8). Information about reader demographics was unavailable in the publicly available documentation examined for this investigation. Since WOMAC pain was self-reported, the principal outcome did not depend upon evaluation by an assessor.

For each fold in the outer cross-validation training set, average expected baseline pain was estimated as the average baseline WOMAC pain amongst training knees in each categorical KL grade group. Discordance was defined as the difference between actual baseline WOMAC pain and expected training-set baseline pain within the same KL grade category, yielding positive and negative values representing more/less pain than expectation. The primary exposure of interest was the absolute value of discordance.

Primary prediction model and internal validation

For prediction modeling, we utilized elastic-net regression as our core modeling technique. The elastic net is an interpretable linear model which regularizes correlated baseline covariates without performing outcomes-based feature selection. Continuous covariates were imputed based on medians in the training folds alone using missingness indicators, then standardized within the training data. Categorical covariates were coded explicitly to include a Missing category, and dummy-encoded. Unknown categories were ignored during prediction. All covariates were processed in an identical fashion regardless of participant sociodemographic characteristics.

A nested five-fold cross-validation procedure was performed at the participant grouping level, using a fixed seed of 20260721. The outer CV procedure produced exactly one held-out prediction for each eligible knee. Within each outer training fold, the elastic net's penalty parameter $\alpha \in \{0.001, 0.003, 0.01, 0.03, 0.1, 0.3, 1.0\}$ and L1 mixing ratio $\in \{0.05, 0.25, 0.50, 0.75, 0.95\}$ values were tuned using inner five-fold CV through minimization of MAE. The optimal α was 0.001 for all outer folds (smallest possible value in the parameter range), with the L1 mixing ratio equal to 0.95 for four outer

folds and 0.05 for one fold. Therefore, the amount of regularization was near zero, leading to instability in estimating the mixing proportion. Thus, the resulting fitted elastic-net models bore close resemblance to unregularized linear regression. No preprocessing, discordance calculation, hyperparameter tuning, nor model fitting occurred outside of the training fold data.

Prediction accuracy measures considered in the current report were mean absolute error (MAE), root-mean-square error (RMSE), average bias, calibration intercept, and calibration slope. 95% confidence intervals for the primary measures were calculated through 1,000 participant-resampled bootstraps. Resample sizes varied across analyses depending upon their specific purpose, where descriptive intervals for the longitudinal quartiles required 2,000 resamples, while attrition rate and subgroup statistics calculations called for 5,000 resamples. Resampling was done according to participants rather than knees for the former; participant bootstrapping was defined for the latter.

Development and internal validation were both performed on the same osteoarthritis initiative (OAI) cohort in the exact healthcare setting using identical criteria for participant eligibility, outcome ascertainment, and covariate definition. No external validation cohort was included here. No center-specific variation analyses or updates were conducted. No calibration updates or subgroup-specific refits took place after model freezing.

As described above, our internal cross-validation produced five fold-specific fitted models rather than creating one final production model trained over the entire sample set. Hence, no refitting process was performed once the analysis period concluded. Our technical specification document archives all aspects of the frozen modeling process, including source code and fitted model summary tables, without disclosure of outer-fold participant groupings. Re-running the frozen technical package produces bit-for-bit identical outer-fold partitions and predicted outcomes. For our reproducibility test, we used Python version 3.13.5 along with scikit-learn 1.8.0, statsmodels 0.14.6, and XGBoost 3.1.3.

Primary hypothesis test

The absolute prediction error for each hold-out knee was calculated based on the absolute difference between the WOMAC pain score observed after 48 months and that predicted by our models. We tested our hypothesis with an ordinary least squares regression of the absolute error term on the cross-fitted absolute baseline discordance measure using CR1 clustered sandwich covariance estimator by participants. Our chosen significance threshold was $\alpha = 0.05$. As a prespecified robustness check, the primary parameter estimate was computed with 1,000 bootstrap draws based on individual-level resampling. Quartiles of baseline discordance were utilized only for illustrative purposes here.

Post-freeze baseline-pain sensitivity analysis

Upon completion of this process, we undertook another sensitivity analysis prompted by the internal peer review prior to manuscript submission. The analysis made no changes to any aspects of model fitting, predictions on held-out data, or any previously reported frozen prespecified results. For the first model of the sensitivity analysis, we correlated the magnitude of cross-fitted discordance with baseline WOMAC pain score. Next, the magnitude of held-out prediction error was jointly modeled against both the magnitude of discordance and baseline WOMAC pain score, making use of the same participant-clustered covariance matrix estimator used in our principal analysis. A second model additionally included categorical KL-grade fixed effects.

Longitudinal analysis

Secondary longitudinal analysis included a sample of 7,807 knees with WOMAC pain values measured at baseline and at 24, 36, and 48 months. Elastic-net prediction models were trained separately for each of the three follow-ups using the same baseline predictors, preprocessing methods, and outer participant folds as in the primary analysis. Within each outer training fold, hyperparameter selection was achieved by participant-based five-fold inner cross-validation, while baseline discordances were re-calculated within each training fold.

The resultant absolute prediction error values from 24, 36, and 48 months are arranged in long format and submitted to linear mixed effects modeling with absolute baseline discordance, visit (as a categorical predictor), and interaction terms of discordance-by-visit as fixed effects. Participant-specific random intercepts and knee-specific random intercepts nested within participants are included in the model. The visit at 24 months serves as a reference group. Estimates of discordance-error slopes at each timepoint were calculated using linear combinations of parameters. Modeling is done by maximum likelihood (not restricted maximum likelihood), utilizing the L-BFGS optimization algorithm. Descriptive longitudinal quartile confidence intervals are obtained through bootstrapping using 2,000 replicates sampled at the participant level.

Missing outcome and attrition analysis

Outcome WOMAC scores after 48 months were not imputed. In the main analysis, the sample included only those knees whose 48-month pain outcome could be observed. However, as mentioned above, the knees with unobserved outcomes remained part of a prespecified attrition analysis. Continuous covariates were assessed in terms of difference in means and standard deviations, while discrete covariates used frequency counts. Effect sizes were calculated as standardized mean differences. Differences and 95% confidence intervals were estimated by bootstrapping (n=5,000 iterations) using pairs of knees nested within participants.

Uncertainty estimation and selective abstention

Firstly, it bears mentioning that we estimated absolute prediction error of the primary prediction without including any results from the held-out outer test sets. For each outer training set, we generated predictions outside the folds of the primary model using the selected primary hyperparameters and then used these absolute prediction errors as targets for a second regularized error model predicting error. The inputs to the uncertainty model were prespecified baseline covariates, primary-model prediction, estimated expected baseline pain using training samples, signed discordance, and discordance absolute value. Using this method, the uncertainty model estimated error scores for knees in the independent outer-test set. As the latter has already been discussed previously, we emphasize here that because discordance has been used as one of the predictors in this approach, any correlation between uncertainty and error or subgroup-wise deferral behavior does not provide additional validation for a discordance effect beyond the initial findings.

We ranked the predictions based on their estimated uncertainties from smallest to largest. The performance metrics were repeated after deferring the top 5%, 10%, 15%, and 20% uncertain predictions. The predefined threshold of abstention was set to be 10%. Risk and coverage analysis evaluated coverage as the proportion of knees having predictions, whereas risk was evaluated as the prediction error for those retaining predictions. The comparison of targeted deferral versus random deferral was conducted at identical coverages, utilizing 1,000 simulations of random knee-level deletion

at each coverage rate. Finally, uncertainty was correlated with absolute prediction error using Spearman correlation coefficient.

Subgroup audit

For the primary analysis cohort, the frozen 10% global abstention threshold was applied universally, without recomputation of subgroup thresholds. Metrics including error, bias, calibration, abstention rate, coverage, and MAE of retained cases were tabulated stratified by sex, race, Hispanic origin, baseline Kellgren-Lawrence (KL) grade, history of knee injury, and knee surgery. Ninety-five percent confidence intervals were calculated via participant-level bootstrap sampling with 5,000 replicates. For metrics other than calibration, these calculations were performed at the level where there were at least 50 knees, whereas calibration was computed only for levels above 100 knees. These analyses were exploratory in nature and did not affect the primary model or abstention threshold. Subgroup fairness constraints, demographic weighting, subgroup calibration, or subgroup abstention threshold was not used for any purposes; fairness was considered through subgroup reporting alone.

Sensitivity analyses and knee-replacement audit

Seven prespecified sensitivity analyses examined whether discordance and error were affected by selection biases induced by how the cohort, outcome, discordance variable, missing data management, presence of bilateral knees, and/or modeling strategy were defined. In these sensitivity analyses, the cohort had Kellgren-Lawrence (KL) grade 2+ osteoarthritis at baseline, positive and negative discordances were separately analyzed, predictions were made at 48 months of WOMAC pain change, expected pain estimates were re-calculated using baseline covariates, only complete-case data was considered, a single knee was selected for each individual, and XGBoost (XGBRegressor) replaced elastic-net without changing the grouped cross-validation scheme. For the covariate-adjusted sensitivity analysis estimating expected pain, covariates were adjusted in the model used to estimate expected pain but were not adjusted in the prediction-error regression of absolute held-out error.

The prespecified secondary knee-replacement outcome was audited before modeling. Despite having covariates for side-specific event time indicating total knee replacement surgery, derivation could not produce covariates indicating partial-knee replacement surgery, nor did it contain censoring variables indicating survival/death and loss to follow-up time. Therefore, the time-to-event model was not fitted.

Reporting transparency and open science

Our reporting was audited against the TRIPOD+AI Guideline and its associated abstract checklist by Collins et al. The internal preregistration of our protocol occurred before testing our primary discordance-error hypothesis, but the study was not registered in a public registry. The OAI source data are accessible via the NIMH Data Archive (NDA) OAI resource, and therefore we abide by its data access guidelines. Our locked-down frozen analytical codebase, analysis parameters, tuning summaries, and aggregate results remain privately stored in our technical package. Row-level files containing participant folds and additional derivations from the OAI are stored privately without intent of public redistribution. Preparation is underway for our public code repository, which will omit participant-level OAI data. No patients or general public participants contributed to this methodology paper.

Results

Cohort characteristics and descriptive findings

Our final baseline sample included 8,920 knees belonging to 4,490 patients (Table 1 and Figure 1). Our sample had an average age of 61.11 years (SD = 9.17), mean body mass index (BMI) of 28.58 kg/m² (SD = 4.82), and 58.1% knees belonging to females. Average baseline pain score as assessed using WOMAC pain scale was 2.34 (SD = 3.26). But a substantial floor effect was present, with 3,946 knees (44.2%) reporting a baseline pain score of zero. The distribution for KL Grades 0, 1, 2, 3, and 4 was 38.6%, 17.8%, 26.5%, 13.9%, and 3.3%, respectively. There was variation in baseline pain according to KL Grade, but a clear trend towards increasing baseline pain with radiographic severity was evident (Figure 2).

688 knees lacked data on our primary outcome measure, WOMAC pain at 48 months post-baseline. Knees without 48-month WOMAC pain had higher baseline pain scores compared to those knees with available 48-month WOMAC pain scores (difference = 1.18; 95% CI, 0.83-1.54; standardized mean difference (SMD) = 0.33) and higher baseline BMIs (difference = 0.98; 95% CI, 0.41-1.57; SMD = 0.20). A greater proportion of the former group consisted of individuals reported as Black or African American race (31.7% vs. 16.2%), while a lesser proportion was Caucasian (63.8% vs. 81.6%). KL Grade ≥2 knees were also more frequent in the former group (48.3% vs. 43.3%). These imbalances indicate possible follow-up selection bias.

Table 1: Baseline characteristics of the OAI knee cohort

Characteristic	Mean (SD) or n (%)	Median [IQR] / denominator	Missing n
Knees, n	8,920		0
Participants, n	4,490		
Age, years	61.11 (9.17)	61.0 [53.0, 69.0]	0
BMI, kg/m²	28.58 (4.82)	28.2 [25.0, 31.7]	8
Baseline WOMAC pain (0-20)	2.34 (3.26)	1.0 [0.0, 4.0]	0
Baseline WOMAC pain = 0	3,946 (44.2%)		0
Female sex	5,182 (58.1%)	N=8,920	0
Race: White or Caucasian	7,148 (80.2%)	N=8,912	8
Race: Black or African American	1,550 (17.4%)	N=8,912	8
Race: Asian	76 (0.9%)	N=8,912	8
Race: Other Non-white	138 (1.5%)	N=8,912	8
Hispanic ethnicity	106 (1.2%)	N=8,918	2
KL grade 0	3,440 (38.6%)	N=8,920	0
KL grade 1	1,587 (17.8%)	N=8,920	0
KL grade 2	2,362 (26.5%)	N=8,920	0

Characteristic	Mean (SD) or n (%)	Median [IQR] / denominator	Missing n
KL grade 3	1,237 (13.9%)	N=8,920	0
KL grade 4	294 (3.3%)	N=8,920	0
Any medial JSN (grade ≥1)	3,137 (35.2%)	N=8,920	0
Any lateral JSN (grade ≥1)	728 (8.2%)	N=8,920	0
Prior knee injury	2,385 (27.0%)	N=8,832	88
Prior knee surgery	1,105 (12.4%)	N=8,903	17
Observed 48-month WOMAC outcome	8,232 (92.3%)	N=8,920	688

Note: Numbers are expressed as means (standard deviations), medians [interquartile range], or numbers (%), as specified. Calculations of percentages use available case denominators. Patient demographics are reported at the knee-record level. Abbreviations are BMI (body mass index), JSN (joint-space narrowing), KL (Kellgren-Lawrence score), and WOMAC (Western Ontario and McMaster Universities Osteoarthritis Index). At 48 months' follow-up, n = 688 missing out of 8,920 initial knees.

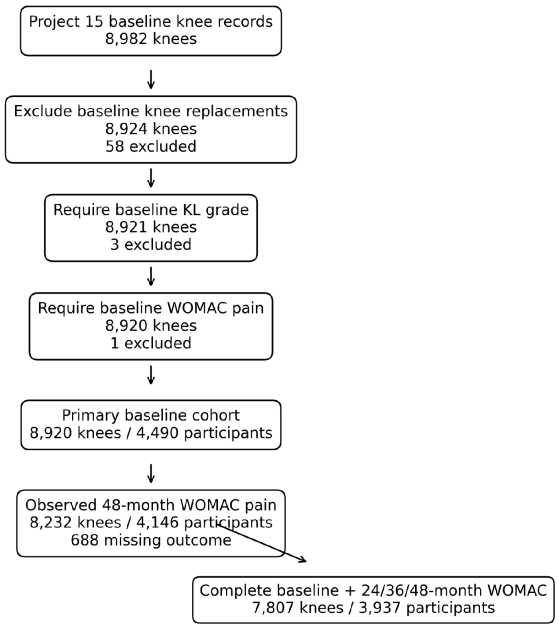


Fig 1: Establishment of cohorts for the primary and longitudinal investigations. For the primary investigation, the baseline cohort consisted of 8,920 knees belonging to 4,490 individuals. Specifically, 8,232 knees from 4,146 individuals had available WOMAC pain scores at 48 months, whereas 7,807 knees from 3,937 individuals had fully available longitudinal WOMAC pain data spanning 48 months.

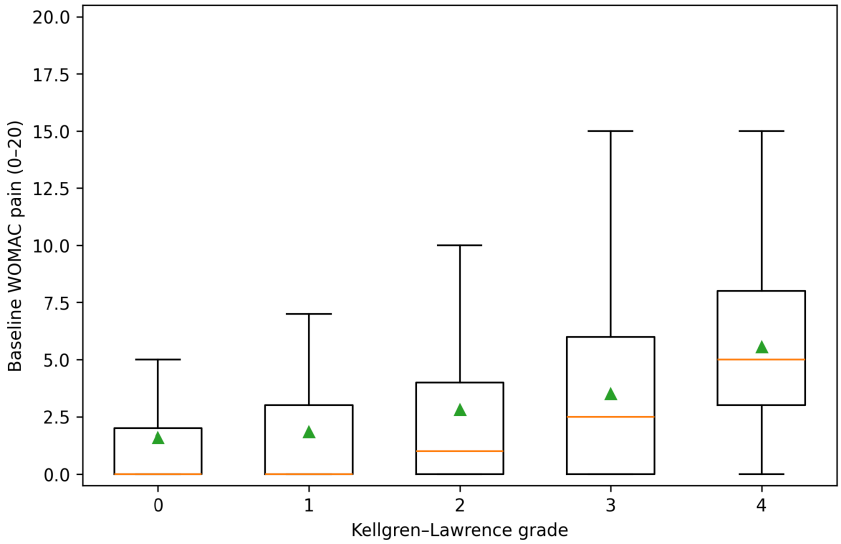


Fig 2: Pain from WOMAC stratified by Kellgren-Lawrence Grade. Boxes show the interquartile range; the horizontal line indicates the median; whiskers denote the non-outlying range. Group means are indicated by green triangles. Individual outliers were removed from visualization but kept in the statistical analyses.

Primary model performance and discordance hypothesis

Nested five-fold participant-grouped cross-validation resulted in exactly one prediction being held-out for each of the 8,232 knees from 4,146 participants. Checks verified that no participant straddled an outer training-test split border, no follow-up outcomes were used as baseline features, and exactly one prediction was produced for each eligible knee. Running the pipeline frozen produced outer folds with bitwise identical assignments and bitwise identical predictions as a previous trial run. Our model generated the following metrics on held-out data: held-out MAE of 1.787 WOMAC points (95% CI, 1.741-1.835), RMSE of 2.567 points (95% CI, 2.495-2.645), prediction bias of 0.000 points (95% CI, -0.065 to 0.061), calibration intercept of 0.019 (95% CI, -0.076 to 0.115), and calibration slope of 0.991 (95% CI, 0.936-1.045). These metrics are summarized in Table 2.

For each increase in cross-fitted absolute discrepancy between structure and pain by one WOMAC point in the frozen unadjusted primary analysis, there was an associated increase in absolute prediction error of 0.301 (participant-clustered 95% CI, 0.275 to 0.326, $p = 8.18\times10^{-118}$). Similar results were found using participant-level bootstrapping to produce a similar 95% CI of 0.277 to 0.326. Absolute errors for the four Qs ranged from 1.053 to 2.636 points (Figure 3):

- Q1: 1.874
- Q2: 1.053
- Q3: 1.587
- Q4: 2.636

Clearly, these results are non-monotonic. The unexpectedly low error value for Q2 results from folding of the exposure and low floor for baseline pain: 86.2% of knees in Q2 had 0 baseline pain score compared to 0% for Q1, 56.5% for Q3, and 37.5% for Q4. The quartiles therefore should not be interpreted as a stepwise dose-response scale.

Post-freeze baseline-pain adjustment

Strong correlations were observed between absolute disagreement and baseline WOMAC pain scores (Pearson r = 0.751). Pain score alone explained more variance in absolute error of prediction compared to disagreement alone (R^2 = 0.202 vs 0.111). Disagreement had a coefficient of -0.011 (95% CI, -0.043 to 0.022; p = 0.52) when controlling for baseline WOMAC pain, whereas the coefficient for baseline pain was 0.265 (95% CI, 0.246-0.284). Addition of KL-grade fixed effects made the coefficient for disagreement more negative, to -0.020 (95% CI, -0.053 to 0.012; p = 0.22) while decreasing the baseline pain coefficient to 0.249 (95% CI, 0.231-0.268). As a result, the unadjusted discordance association did not separate from baseline symptom burden in this added analysis.

Table 2: Primary model performance, adjusted sensitivity, and uncertainty results

Measure	Estimate	95% CI / interval	p value / note
Held-out MAE	1.787	1.741 to 1.835	
Held-out RMSE	2.567	2.495 to 2.645	
Mean prediction bias	0.000	-0.065 to 0.061	
Calibration intercept	0.019	-0.076 to 0.115	
Calibration slope	0.991	0.936 to 1.045	
Unadjusted discordance → absolute error β	0.301	0.275 to 0.326	$p = 8.18 \times 10^{-118}$
Discordance β adjusted for baseline pain*	-0.011	-0.043 to 0.022	p = 0.52
Discordance β adjusted for baseline pain + KL*	-0.020	-0.053 to 0.012	p = 0.22
Uncertainty vs absolute error (Spearman ρ)	0.573		p < 0.001
Retained-case MAE after 10% targeted abstention	1.582	1.540 to 1.625	823 knees deferred
Relative MAE reduction after 10% targeted abstention	11.5%		
Random 10% abstention MAE	1.788	1.774 to 1.801	95% random-abstention interval

*Sensitivity analysis post-freezing has been conducted internally and does not affect model fitting or freezing. The metrics utilized to predict performance and abstention ranges include the mean absolute error (MAE) and root mean square error (RMSE), with their primary confidence intervals generated through participant-level bootstrap sampling. Kellgren-Lawrence scoring is abbreviated KL.

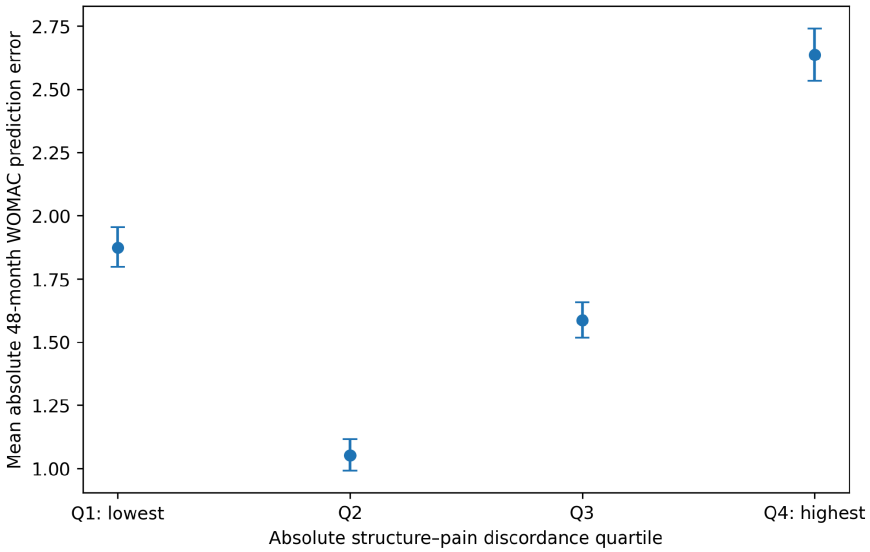


Fig 3: The graph depicts 48 month absolute prediction error of held-out cases by quartiles of structure-pain discordance at baseline. Each dot represents the mean absolute prediction error of WOMAC scores, with accompanying 95% bootstrap confidence intervals based on resampled individuals. Note that the quartile groupings are merely descriptive since the primary hypothesis was tested using a continuous exposure.

Longitudinal prediction

The longitudinal analysis included 7,807 knees belonging to 3,937 patients and resulted in 23,421 leakage-protected held-out predictions at 24, 36, and 48 months. The values for MAE were 1.610, 1.683, and 1.755 WOMAC units, respectively, while the slopes of calibration tended toward unity. Baseline discordance correlated positively with prediction error at each visit timepoint in the frozen analysis. The coefficient estimates were 0.236 (95% CI, 0.216 to 0.255) for 24 months, 0.270 (95% CI, 0.251 to 0.290) for 36 months, and 0.257 (95% CI, 0.238 to 0.276) for 48 months; p-values in each case were less than 0.001 (Table 3). The highest-discordant quartile showed the greatest mean absolute error at each visit time point, though there was not always an obvious monotonically increasing relationship for the four-quartiles trendline. It should be noted that the values for the longitudinal 48-month visit differ somewhat from those for the primary MAE of 1.787 and slope of 0.301 because the longitudinal analysis required complete pain observations at all three follow-ups and therefore used 7,807 rather than 8,232 knees.

Table 3: Longitudinal prediction results in the complete-trajectory cohort

Visit month	Knees / participants	MAE	Calibration slope	Discordance-error slope (95% CI); p
24	7,807 / 3,937	1.610	0.994	0.236 (0.216 to 0.255); p < 0.001
36	7,807 / 3,937	1.683	0.991	0.270 (0.251 to 0.290); p < 0.001
48	7,807 / 3,937	1.755	0.989	0.257 (0.238 to 0.276); p < 0.001

Note: Each of the three iterations uses the identical full-trajectory cohort comprising 7,807 knees from 3,937 individuals. Hence, the 48-month estimates differ numerically from those found in the main analysis of 8,232 knees. WOMAC = Western Ontario and McMaster Universities Osteoarthritis Index; MAE = Mean Absolute Error.

Uncertainty and selective abstention

An additional iteration returned the full suite of uncertainty scores and risk coverage values. Estimated uncertainty exhibited a positive correlation with held-out absolute prediction error (Spearman $\rho = 0.573$; $p < 0.001$). Removing the top-scoring 823 knees in terms of uncertainty (10.0% of 8,232) lowered the average held-out MAE from 1.787 to 1.582 (95% CI, 1.540-1.625), representing an absolute improvement of 0.205 points and a relative improvement of 11.5%. A random selection process whereby 10% of predictions were removed resulted in an MAE of 1.788 and 95% confidence interval for random abstention of 1.774 to 1.801; the latter did not overlap the former targeted retention CI. Targeted removal of 5%, 15%, and 20% led to MAEs of 1.669, 1.504, and 1.437, respectively (Fig. 4; Table 2). Since both signed and absolute disagreement are included predictors in the uncertainty system, the above results suggest that the estimated uncertainty system can focus errors despite not being a replication of the signed disagreement result.

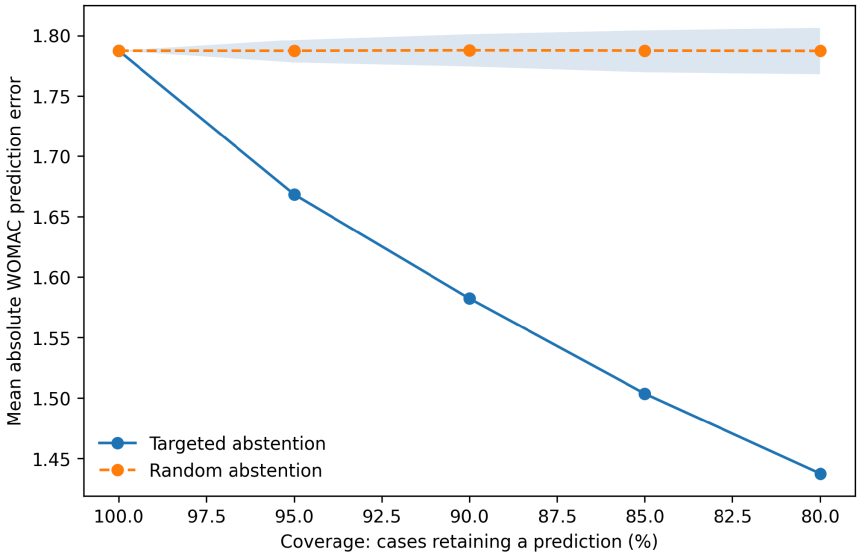


Fig 4: Risk-coverage analysis for selective abstention. Here, the selected target line shows the average absolute error for kept instances following exclusion of instances with highest uncertainty scores based upon nesting (i.e., selecting from within each group of same coverage). Meanwhile, the random-abstention dashed line shows the mean of 1,000 randomly removed knees per matched coverage.

Subgroup coverage and performance

Predictive accuracy and abstention differed descriptively among various subgroups stratified by demographic characteristics and medical histories. For instance, female knees had an MAE of 1.868 and an abstention rate of 11.9% while male knees had an MAE of 1.675 and an abstention rate of 7.3%. White/Caucasian-recorded knees had an MAE of 1.597 and an abstention rate of 5.5% while Black/African American-recorded knees had an MAE of 2.715 and an abstention rate of 32.2%. Higher abstention rates were also observed according to higher radiographic baseline severity, increasing from a 2.9% abstention rate in Kellgren-Lawrence (KL) category 0 to 39.5% in KL category 4 knees. Knees

with previous injuries and knees with previous surgeries had higher errors and abstentions as well when compared to knees without these histories (see Figure 5 and Table 4).

Within the set of subgroup levels consisting of ≥100 knees, calibration slopes were estimated in the range [0.842, 1.066], with 95% CIs encompassing 1. Use of the frozen abstention threshold improved the MAE among all retained cases for all subgroups comprising ≥100 knees. One exception to this pattern was the small "Asian" subgroup which had an increase in MAE after abstention but with an uncertain estimation due to imprecision. This discrepancy should not be viewed causally as indicating any form of racial or ethnic bias.

Table 4: Subgroup error, abstention, and retained-case performance under the frozen global 10% rule

Dimension	Level	Knees / participants	MAE (95% CI)	Abstention % (95% CI)	Retained MAE (95% CI)
Sex	Female	4795 / 2415	1.868 (1.805-1.93)	11.9% (10.9-13.0)	1.635 (1.579-1.69)
Sex	Male	3437 / 1731	1.675 (1.605-1.74)	7.3% (6.3-8.3)	1.512 (1.452-1.57)
Race	White or Caucasian	6709 / 3379	1.597 (1.552-1.64)	5.5% (4.9-6.1)	1.481 (1.439-1.52)
Race	Black or African American	1332 / 671	2.715 (2.573-2.86)	32.2% (29.3-35.1)	2.236 (2.092-2.38)
Race	Asian	65 / 33	1.588 (1.125-2.17)	6.2% (0.0-13.8)	1.610 (1.119-2.24)
Race	Other Non-white	118 / 59	2.190 (1.757-2.68)	16.1% (8.5-24.6)	2.014 (1.536-2.58)
Hispanic ethnicity	No	8138 / 4099	1.782 (1.735-1.83)	10.0% (9.2-10.7)	1.578 (1.536-1.62)
Hispanic ethnicity	Yes	92 / 46	2.181 (1.676-2.76)	13.0% (5.4-21.7)	1.958 (1.451-2.56)
Baseline KL grade	0	3181 / 2036	1.345 (1.283-1.40)	2.9% (2.3-3.6)	1.260 (1.203-1.31)
Baseline KL grade	1	1490 / 1210	1.599 (1.506-1.69)	3.8% (2.8-4.9)	1.514 (1.424-1.60)
Baseline KL grade	2	2184 / 1655	2.056 (1.971-2.13)	13.7% (12.1-15.4)	1.814 (1.736-1.89)
Baseline KL grade	3	1111 / 897	2.508 (2.381-2.64)	24.1% (21.4-26.8)	2.161 (2.038-2.28)
Baseline KL grade	4	266 / 265	2.910 (2.665-3.16)	39.5% (33.5-45.5)	2.614 (2.334-2.91)
Prior knee injury	No	5963 / 3672	1.679 (1.627-1.73)	7.9% (7.1-8.7)	1.506 (1.459-1.55)

Dimension	Level	Knees / participants	MAE (95% CI)	Abstention % (95% CI)	Retained MAE (95% CI)
Prior knee injury	Yes	2191 / 1760	2.085 (2.002-2.17)	15.7% (14.1-17.4)	1.811 (1.734-1.89)
Prior knee surgery	No	7215 / 4013	1.742 (1.692-1.79)	8.9% (8.2-9.7)	1.550 (1.504-1.59)
Prior knee surgery	Yes	1003 / 878	2.114 (2.000-2.23)	17.6% (15.3-20.1)	1.845 (1.737-1.95)

Note: The confidence intervals were computed via bootstrapping from 5,000 repetitions. MAE stands for mean absolute error, whereas KL signifies Kellgren-Lawrence grades. The numbers of participants are unique for each row. The totals for KL scores, injury statuses, and surgical interventions can add up to greater than 4,146 because a single participant might provide both knees with their respective statuses.

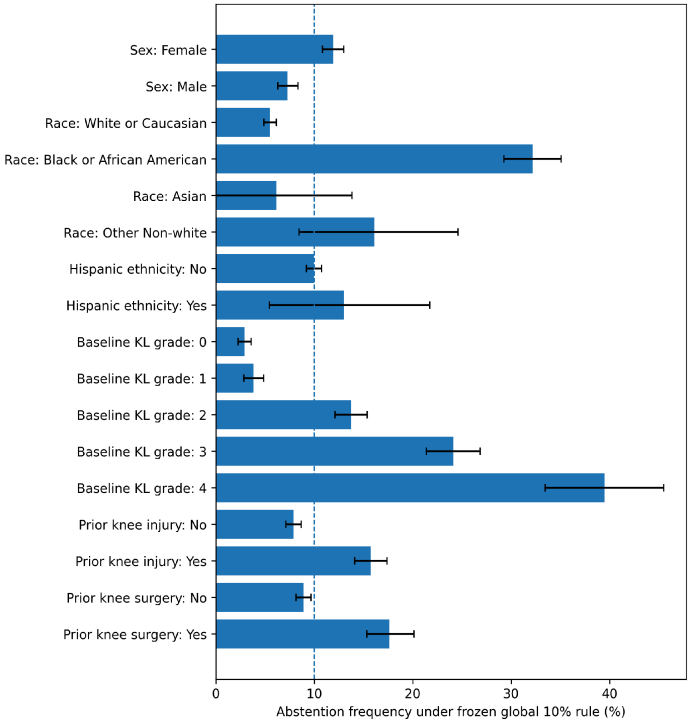


Fig 5: The percentages of abstainers per subgroup under the global 10% abstainer threshold restriction are given. Here, the vertical line shows the overall 10% abstainer goal threshold. Bootstrap confidence intervals of 95% using participants are displayed using error bars. Coverage differences alone do not imply demographic biases.

Sensitivity analyses and knee-replacement feasibility

We conducted seven prespecified sensitivity analyses on discordance as a predictor of error (Table 5). Limiting the sample to KL Grade ≥2 yielded an estimated coefficient of 0.204 (95% CI, 0.167-0.241); forecasting WOMAC 48-month change gave 0.329 (95% CI, 0.298-0.361); the complete-case analysis yielded 0.305 (95% CI, 0.279-0.331); and one-knee-per-participant gave 0.306 (95% CI, 0.272-0.340). The covariate-adjusted-discordance analysis with S4 changed the training-fold pain model but not the prediction-error regression model. Since we tested the impact of covariate

adjustment on pain at baseline in post-freeze S8, this analysis did not directly address baseline pain confounding.

Our signed-analysis results revealed asymmetry in how positive vs negative discordances influenced error. For knees exhibiting more pain than expected, each unit increase in positive signed discordance was linked to an extra 0.284-unit increment in absolute error (95% CI, 0.259-0.310). Conversely, larger discordances toward less pain yielded smaller errors (slope coefficient, -0.110; 95% CI, -0.153 to -0.066; Wald $p < 0.001$). Our findings arise from a combination of the pain-floor constraint on the baseline covariates and the folding property of absolute discordance in estimating deviation from a KL-grade expected value. One must resist reading causal distinctions into the asymmetric behavior observed here. Overall, gradient-boosting tree model performance was comparable to elastic-net regression: mean absolute error was 1.791 compared to 1.787, with bootstrap difference of 0.004 (95% CI, -0.004 to 0.012). The relationship between discordance and error remained similar using gradient boosting (coefficient estimate, 0.298; 95% CI, 0.269-0.326).

The replacement audit resulted in the identification of 885 cases classified as confirmed follow-up total knee replacements. Another 29 cases having status code 1 had a replacement day value recorded, giving a total of 914 cases with a replacement-day value. These 29 were not considered extra confirmed event cases. Since no valid censoring times for non-event knees could be extracted from the stored derived dataset, the pre-specified time-to-event analysis could not be completed.

Table 5: Sensitivity analyses

Analysis	Knees / participants	Effect estimate	95% CI	Technical note
S1 KL ≥ 2 restriction	3561 / 2332	0.204	0.167 to 0.241	$p < 0.001$
S2a positive signed discordance	8232 / 4146	0.284	0.259 to 0.310	Directional slope
S2b low-pain discordance magnitude	8232 / 4146	-0.110	-0.153 to -0.066	Wald $p < 0.001$ vs S2a
S3 48-month WOMAC change outcome	8232 / 4146	0.329	0.298 to 0.361	$p < 0.001$
S4 covariate-adjusted discordance	8232 / 4146	0.319	0.294 to 0.345	$p < 0.001$
S5 complete-case analysis	8127 / 4108	0.305	0.279 to 0.331	$p < 0.001$
S6 one knee per participant	4146 / 4146	0.306	0.272 to 0.340	$p < 0.001$
S7 gradient-boosted-tree comparator	8232 / 4146	Slope 0.298; MAE Δ 0.004	Slope 0.269 to 0.326; Δ -0.004 to 0.012	No material MAE improvement

Analysis	Knees / participants	Effect estimate	95% CI	Technical note
S8a* discordance adjusted for baseline pain	8232 / 4146	-0.011	-0.043 to 0.022	p = 0.52
S8b* discordance adjusted for baseline pain + KL	8232 / 4146	-0.020	-0.053 to 0.012	p = 0.22

Note: Unless otherwise specified, effect estimates correspond to the slope of absolute prediction error associated with a one-unit increase in the particular discordance measure considered. S4 impacts the calculation of the expected value of pain, but has no effect on the error regression using baseline WOMAC pain. S8 was introduced after analysis was locked down following internal review. Its purpose was to correct the error regression and thus had no impact on any pre-specified finding. KL = Kellgren-Lawrence; MAE = mean absolute error; Δ = difference.

Discussion

The prespecified unadjusted primary analysis indicated substantial predictive discordance between absolute structural-pain discordance and the 48-month prediction error. However, this relationship must be reassessed in the context of post-freeze sensitivity analyses. Absolute structure-pain discordance showed strong correlation with the baseline WOMAC pain score ($r = 0.751$). Moreover, the baseline pain term explained substantially more variance in absolute error ($R^2 = 0.202$) than did structure-pain discordance ($R^2 = 0.111$). When controlling for both terms within the same participant-clustered error regression, the discordance coefficient approaches zero and is statistically insignificant. Even accounting for KL fixed effects has little bearing; the discordance term remains non-significant. Therefore, a more appropriate inference might be drawn: greater symptom burden is associated with diminished predictability. Rather than being indicative of a genuine physiological relationship, absolute discordance serves as an inferior predictor compared to baseline pain. Subtracting the crude, pre-calibrated expectation of pain corresponding to each KL grade introduces added noise.

This notion receives empirical support from descriptive quartile analysis and signed sensitivity analysis. Empirically, Q2 consists almost entirely of pain-free KL 0/1 knees; thus, the group displays less average prediction error relative to both Q1 and Q3. Such ordering results in a non-monotonic quartile trend rather than dose-response. Likewise, positive discordance denotes pain above expectations, while increasingly large values of negative discordance approach the WOMAC floor score. No separate biological explanation is required. Rather, this shows that absolute discordance did not retain an independent association with prediction error after baseline pain adjustment in these data.

Uncertainty analysis answers yet another practical question. Using the nested error regression technique to estimate uncertainty and defer high-uncertainty cases resulted in MAE improvement from 1.787 to 1.582 WOMAC points, having trimmed away 823 cases. Furthermore, the targeted uncertainty range does not overlap the random abstention range. However, it is essential to recognize that signed and absolute discordance were integral predictors within the uncertainty model itself. Therefore, the

correlation between uncertainty and error cannot be cited as corroborative evidence in favor of discordance effects. Subgroup analysis highlights this limitation: abstention ranged from 2.9% for KL 0-grade knees to 39.5% at KL 4, varying according to demographics and clinical factors. As such, selective prediction should be accompanied by an abstention rate.

1.787 WOMAC points constitute the internal mean absolute error (MAE) when holding out a test sample of predictions. Direct ranking against other knee-pain models cannot occur due to disparate pain outcomes, measurement scales, patient populations, forecast horizons, and validation procedures described in the relevant literature (Tong et al. 1048). Even within the present analysis, findings regarding regularization must be reconsidered. Tuning yielded a minimum α at each fold iteration. Moreover, one fold demonstrated abrupt transition in the mixing ratio of L1 regularization. In summary, fitted linear models approximate ordinary least squares. Gradient boosting produced no material reduction in the internal test set MAE.

None of these results warrant automation of decision making, therapeutic recommendations, or patient-specific prediction. Internal cross-validation occurred solely within the Osteoarthritis Initiative dataset, without evaluation of external populations, clinical workflow, or interface. Practical implications are modest: baseline symptom severity correlates positively with mean prediction error, while the nested uncertainty model allows identification and exclusion of a subset of cases where errors are driven by forecast unreliability. External validation is required to confirm these patterns persist in other data.

Limitations

It is worth discussing several limitations in this study. First, although the discordance analysis itself was prespecified, the adjustment for the baseline WOMAC pain level was added only after freezing the analysis pipeline. Thus, the post-freeze adjustment serves merely as a sensitivity analysis of the initial finding rather than a secondary confirmation test. Specifically, absolute discordance shows extremely strong correlation with baseline pain levels, meaning that the unadjusted relationship should not be taken as evidence for an effect independent of the symptom severity.

Second, estimated expected pain is based solely on the mean WOMAC pain in each of the five categorical KL grades, making the conditioning function a crude five-bin step-function. Even the KL grade itself is an artificially coarse grouping of the radiographic characteristics to summarize radiographs, ignoring synovitis, bone marrow lesions, meniscal abnormalities, sensitization, psychosocial factors, and other possible mediators of pain generation. Although the S4 expected pain sensitivity analysis did account for other covariates, it did so in a way that modifies discordance construction instead of error regression.

Third, 7.7% of baseline knees lack data at the 48-month endpoint. The attrition analysis finds descriptive differences in baseline pain scores, BMI, and the racial makeup. Some subjects contribute both knees for the analysis; while we took into account dependence by using participant group folds, clustered confidence intervals, and participant bootstrap resampling, the two knees per person are still physiologically correlated. Treatments undergone during follow-up are not included.

Fourth, validation is performed internally within the OAI dataset. Multiple executions serve to illustrate deterministic nature of the frozen analysis pipeline instead of replication on a fresh dataset. Center-level heterogeneity was not assessed individually. Race was included as a predictor in the model and assessed for subgroup differences in its performance. But no causal inference regarding the biology

behind observed differences in performance by race subgroups can be drawn. Partial-replacement and non-event censoring fields were unavailable in the derived dataset.

Finally, the prespecified time-to-event analysis of joint replacement could not be carried out since the derived dataset did not contain partial replacements nor censoring information. Therefore, current conclusions concern only prediction and deferral of WOMAC pain trajectory over follow-up time.

Conclusion

The results presented herein indicate that there is an association between absolute discordance between structure and pain and 48-month prediction error in the frozen unadjusted primary analysis, but this association did not persist after post-freeze adjustment for WOMAC pain at baseline. As such, symptom load at baseline rather than discordance itself appears to be the stronger candidate predictor of predictability failure in these data. Deferral based on uncertainty did reduce error and decrease retained-case mean absolute error (MAE), although there were differences by subgroup. Uncertainty estimation included discordance as well. The model should undergo external validation before clinical implementation or transportability inference.

Declarations

Author contributions

Sricharan Karthigeyan: Conceptualization; Data curation; Investigation; Methodology; Project administration; Writing - original draft; Writing - review & editing. Yuheng Tang: Formal analysis; Methodology; Software; Visualization; Writing - review & editing.

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Competing interests

The authors declare no competing interests.

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Transparency and data access

Ethics and data access

Ethical clearance for the OAI study was obtained by each clinical site and coordinating center where the study took place. All individuals participated under written informed consent, in compliance with the OAI cohort protocol (Nevitt et al.). Our current analysis involves the use of previously collected, anonymized OAI data that were retrieved from the National Institute of Mental Health (NIMH) Data Archive (NDA) OAI resource according to the relevant repository and data usage policies. Recruitment of new participants and any engagement with participants were unnecessary. Our models

were developed purely for the purpose of methodological investigation and never applied clinically for diagnosis, surgical recommendation, treatment adjustment, or prediction reporting.

Data availability

Data for source OAI analyses can be obtained through the NIMH Data Archive (NDA) resource (NIMH Data Archive), provided access criteria are fulfilled according to the policies and regulations currently governing use of that repository. The authors have opted against redistributing participant- and/or knee-level derived OAI datasets publicly. Analysis findings have been archived as part of the frozen technical package for Study 3398 (DOI: 10.15154/v7rp-4489).

Code and materials availability

The frozen technical package includes the analysis software codebase, analysis specification documentation, tuning summary reports, and output results datasets. The fold allocation tables, prediction outputs, and other row-level outputs based on the OAI data are retained privately and are not intended for public redistribution. An anonymized public replication repository is currently under preparation without OAI data inclusion.

Generative AI disclosure

ChatGPT (Chatbot based on OpenAI's language model) was utilized exclusively as a debugging tool during the code implementation phase in developing this paper's results. No usage of the AI was made for writing, initial coding or editing any drafts of the eventual publication, which was prepared solely by the corresponding human authors.

Prior dissemination and author consent

This work has not previously been published or posted on another preprint server. Both authors consent to submission of this manuscript to The Curieux Review.

Works Cited

Ahadian, P., et al. "A Discordance-Aware Multimodal Framework with Multi-Agent Clinical Reasoning." *Preprints.org*, 2026, https://doi.org/10.20944/preprints202604.0342.v1. Preprint, not peer reviewed.

Alkhatatbeh, A., et al. "Predicting Long-Term Pain Resilience in Knee Osteoarthritis: An Osteoarthritis Initiative Nomogram." *Bioengineering*, vol. 13, no. 1, 2026, article 96, https://doi.org/10.3390/bioengineering13010096.

Bedson, J., and P. R. Croft. "The Discordance between Clinical and Radiographic Knee Osteoarthritis: A Systematic Search and Summary of the Literature." *BMC Musculoskeletal Disorders*, vol. 9, 2008, article 116, https://doi.org/10.1186/1471-2474-9-116.

Chen, T., and C. K. Or. "Automated Machine Learning-Based Prediction of the Progression of Knee Pain, Functional Decline, and Incidence of Knee Osteoarthritis in Individuals at High Risk of Knee Osteoarthritis: Data from the Osteoarthritis Initiative Study." *Digital Health*, vol. 9, 2023, article 20552076231216419, https://doi.org/10.1177/20552076231216419.

Collins, G. S., et al. "TRIPOD+AI Statement: Updated Guidance for Reporting Clinical Prediction Models That Use Regression or Machine Learning Methods." *BMJ*, vol. 385, 2024, e078378, https://doi.org/10.1136/bmj-2023-078378.

Finan, P. H., et al. "Discordance between Pain and Radiographic Severity in Knee Osteoarthritis: Findings from Quantitative Sensory Testing of Central Sensitization." *Arthritis & Rheumatism*, vol. 65, no. 2, 2013, pp. 363-372, https://doi.org/10.1002/art.34646.

Gao, F., et al. "Uncertainty-Quantified and Explainable Age- and Sex-Aware Contrastive Learning for Knee Osteoarthritis Classification." 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI), IEEE, 2026, pp. 1-5, https://doi.org/10.1109/ISBI61048.2026.11515886.

Guan, B., et al. "Deep Learning Approach to Predict Pain Progression in Knee Osteoarthritis." *Skeletal Radiology*, vol. 51, no. 2, 2022, pp. 363-373, https://doi.org/10.1007/s00256-021-03773-0.

Halilaj, E., et al. "Modeling and Predicting Osteoarthritis Progression: Data from the Osteoarthritis Initiative." *Osteoarthritis and Cartilage*, vol. 26, no. 12, 2018, pp. 1643-1650, https://doi.org/10.1016/j.joca.2018.08.003.

Kompa, B., et al. "Second Opinion Needed: Communicating Uncertainty in Medical Machine Learning." *npj Digital Medicine*, vol. 4, 2021, article 4, https://doi.org/10.1038/s41746-020-00367-3.

Li, X., et al. "Predictive Models of Radiographic Progression and Pain Progression in Patients with Knee Osteoarthritis: Data from the FNIH OA Biomarkers Consortium Project." *Arthritis Research & Therapy*, vol. 26, no. 1, 2024, article 112, https://doi.org/10.1186/s13075-024-03346-1.

McConnell, S., et al. "The Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC): A Review of Its Utility and Measurement Properties." *Arthritis & Rheumatism*, vol. 45, no. 5, 2001, pp. 453-461, https://doi.org/10.1002/1529-0131(200110)45:5%3C453::AID-ART365%3E3.0.CO;2-W.

National Institute of Arthritis and Musculoskeletal and Skin Diseases. "Osteoarthritis." *National Institute of Arthritis and Musculoskeletal and Skin Diseases*, https://www.niams.nih.gov/health-topics/osteoarthritis. Accessed 17 Sept. 2026.

National Institute on Aging. “Osteoarthritis Initiative (OAI).” *National Institute on Aging*, https://www.nia.nih.gov/research/resource/osteoarthritis-initiative-oai. Accessed 17 Sept. 2026.

NIMH Data Archive. “Osteoarthritis Initiative (OAI).” NIMH Data Archive, https://nda.nih.gov/oai/. Accessed 21 Sept. 2026.

Neogi, T., et al. “Association between Radiographic Features of Knee Osteoarthritis and Pain: Results from Two Cohort Studies.” *BMJ*, vol. 339, 2009, b2844, https://doi.org/10.1136/bmj.b2844.

Nevitt, M. C., et al. *The Osteoarthritis Initiative: Protocol for the Cohort Study*. University of California, San Francisco, 2006, https://oai.epi-ucsf.org/datarelease/docs/StudyDesignProtocol.pdf. Accessed 17 Sept. 2026.

Osteoarthritis Initiative. *Central Reading of Knee X-Rays for K-L Grade and Individual Radiographic Features of Knee OA: Appendix A. Project 15: Reader Discrepancies and Adjudication Procedures*. 2013, p. 8, http://oai.epi-ucsf.org/datarelease/forms/kXR_SQ_BU_Descrip.pdf?V01XRKL. Accessed 20 Sept. 2026.

Segal, N. A., et al. “Sex Differences in Osteoarthritis Prevalence, Pain Perception, Physical Function and Therapeutics.” *Osteoarthritis and Cartilage*, vol. 32, no. 9, 2024, pp. 1045-1053, https://doi.org/10.1016/j.joca.2024.04.002.

Swaminathan, A., et al. “Selective Prediction for Extracting Unstructured Clinical Data.” *Journal of the American Medical Informatics Association*, vol. 31, no. 1, 2024, pp. 188-197, https://doi.org/10.1093/jamia/ocad182.

Tong, B., et al. “Clinical Prediction Models for Knee Pain in Patients with Knee Osteoarthritis: A Systematic Review.” *Skeletal Radiology*, vol. 53, no. 6, 2024, pp. 1045-1059, https://doi.org/10.1007/s00256-024-04590-x.

Vina, E. R., et al. “Natural History of Pain and Disability among African-Americans and Whites with or at Risk for Knee Osteoarthritis: A Longitudinal Study.” *Osteoarthritis and Cartilage*, vol. 26, no. 4, 2018, pp. 471-479, https://doi.org/10.1016/j.joca.2018.01.020.

Yu, J., et al. “An Interpretable and Trustworthy AI Framework for Large-Scale Longitudinal Structure-Pain Association Studies Using Data from the Osteoarthritis Initiative (OAI).” arXiv, 2026, arXiv:2606.05357, https://doi.org/10.48550/arXiv.2606.05357. Posted 3 June 2026. Preprint, not peer reviewed.